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Cybersecurity and trust formation in digital payment use behaviour in North India

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Abstract
This study examines key factors associated with digital payment systems use behaviour in India, focusing on effort expectancy, grievance redressal, trust, performance expectancy, perceived cybersecurity risks, and social influence. Additionally, it explores the moderating effects of cybercrime experience, education level, and age on these relationships. Survey data were collected from urban regions in North India, and structural equation modelling was employed to analyse the determinants of use behaviour among active users. The analysis reveals that effort expectancy, grievance redressal, and performance expectancy are positively associated with use behaviour. Trust did not exhibit a significant direct association with use behaviour; however, it is positively associated with performance expectancy and is associated with social influence and perceived cybersecurity risks. Cybercrime experience moderates the association between perceived cybersecurity risks and trust, highlighting heterogeneity in trust-formation pathways across user subgroups. This research offers new perspectives on the relationships among trust, cybersecurity concerns, and digital payment use behaviour. It is underpinned by a unique dataset collected from diverse urban regions in North India, offering a novel perspective on user behaviour and use determinants. The findings underscore the importance of mitigating cybersecurity risks and strengthening trust-related mechanisms to support sustained use, offering practical recommendations for policymakers and digital payment providers.

Keywords Cybercrime, Cybersecurity risks, Trust, Technology use, Digital payment systems, India

1 Introduction

The swift progression of digital technology has revolutionised India's financial sector, positioning digital payment systems as essential for cashless transactions. These systems enable seamless financial operations through platforms such as the Internet, mobile networks, and point-of-sale (POS) terminals, offering efficiency and speed that have supported rapid growth in use [1, 2].

India's digital payment ecosystem has evolved rapidly, with platforms like Unified Payments Interface (UPI) [3], Paytm [4], and Google Pay [5]. UPI is pivotal, integrating multiple banks into a streamlined system for real-time fund transfers [6]. Alongside mobile payments, banking cards—both credit and debit—remain foundational, offering secure and convenient transaction options at physical POS and online platforms [7]. The Aadhaar-enabled Payment System (AEPS) utilises biometric infrastructure to enhance financial inclusion, especially in rural regions, by offering fundamental banking services [7]. Moreover, e-wallets like Paytm Wallet and PhonePe further simplify transactions, gaining popularity due to their ease of use for online and in-store payments [8].

The expansion of digital payments in India has been extraordinary. The Reserve Bank of India's 2023–24 report highlights that UPI accounts for over 75% of digital retail payments, with a 50% annual growth in transaction volume and value [9]. Additionally, platforms like AEPS have significantly advanced financial inclusion, especially in underserved regions [10].

However, alongside this rapid growth, the need to fortify cybersecurity measures has become increasingly critical. The cybercrimes targeting digital payment systems continue to present significant challenges for clients and financial institutions [11]. From January to October 2023, India's financial sector experienced more than 13 lakh cybercrimes, averaging about 4400 incidents per day, highlighting the urgency of improving cybersecurity measures to protect the integrity of digital payments [9].

This research examines associations between perceived cybersecurity risk, effort expectancy, grievance redressal mechanisms, performance expectancy, trust, social influence, and digital payment use behaviour in North India. It also explores how age, education, and cybercrime experience moderate selected relationships, thereby shaping variation in use behaviour. The findings aim to guide digital payment providers and policymakers in supporting sustained usage by enhancing cybersecurity and user trust, thereby facilitating sustainable platform development.

The remainder of the paper is structured as follows: Sect. 2 presents an overview of pertinent literature. The research framework, including the factors examined and the hypotheses formulated, is outlined in Sect. 3. Section 4 explains the research methodology, including the questionnaire design, data collection procedures, and data screening. Section 5 discusses the data analysis and results, presenting descriptive statistics, measurement model outcomes, structural model findings, hypothesis testing, and results for mediation and moderation. The discussion of theoretical contributions and practical implications is examined in Sect. 6, which also includes a dedicated limitations subsection (Sect. 6.3) to contextualise interpretation of the findings. Lastly, Sect. 7 delineates the key findings and further research suggestions.

2 Literature review

2.1 Evolution of digital payment systems in India

Critical events have shaped the development of digital payment systems in India. One of the most significant events was the government's demonetization in 2016, which rendered high-denomination currency worthless, hence accelerating the population's shift towards cashless transactions [7]. Studies during this period identified convenience, accessibility, and trust as key factors influencing digital payment use [12]. Despite this push, cash continues to dominate, highlighting ongoing challenges in fully transitioning to a cashless economy [7].

Furthermore, during the COVID-19 pandemic, the demand for contactless transactions during lockdowns intensified, significantly expediting the implementation of digital payments [13]. Small businesses and vendors widely used platforms like the Unified Payments Interface (UPI) and mobile wallets to sustain their operations, which were bolstered by government promotion [14]. However, usage disparities remain driven by demographic factors. Educated individuals and younger generations are more inclined to embrace digital payments owing to higher digital literacy and comfort with mobile-based technologies [12]. Conversely, older adults face barriers like security concerns and low digital literacy, underscoring the need for targeted interventions to address these gaps [15].

Key challenges persist across technical, social, economic, and legal dimensions. Technical issues, such as privacy and security vulnerabilities, erode trust in these systems [16], while social factors, particularly trust and reputation, significantly influence use behaviour [17]. High costs associated with digital payment platforms and limited digital literacy further hinder usage in underserved communities [18]. Moreover, the absence of clear regulatory frameworks exacerbates user hesitancy, highlighting the need for robust legal and policy measures [19].

2.2 UTAUT model

The unified theory of acceptance and use of technology (UTAUT), developed by [20], is widely utilized to investigate technology acceptance and use. The model integrates determinants, like performance expectancy, social influence, and effort expectancy, making it a robust framework for analysing user behaviour [21, 22]. Empirical studies on digital payment use behaviour in India have identified social influence as the most significant factor [23, 24], followed by effort expectancy [7, 25], performance expectancy [26, 27], and facilitating conditions [7, 28].

Extensions to the UTAUT model have incorporated constructs like trust, anxiety, and grievance redressal to better capture the nuances of user behaviour in digital payment use contexts. For example, prior work on mobile payment usage highlighted the role of grievance redressal in relation to use behaviour [25], while perceived credibility has been reported as a substantial predictor of behavioural intention [29].

3 Research model and hypothesis development

We propose a model that extends the UTAUT framework by integrating additional constructs, including perceived cybersecurity risks, trust, and grievance redressal. Furthermore, the model considers moderating factors, including age, education level, and cybercrime experience. Figure 1 depicts the proposed research model, with a comprehensive discussion of each construct provided in the following subsections.

3.1 Perceived cybersecurity risks

The notion of perceived risk, introduced in 1960 to delineate client behaviour and decision-making [30], has evolved with the rise of online transactions. In this study, Perceived Cybersecurity Risks (PCR) pertains to the client's perception of the potential hazards and threats like fraud, unauthorized access, and data breaches associated with digital payment systems.

Prior research indicates that higher perceived risk is associated with lower engagement and willingness to use digital payment systems [31–35]. In the Indian context, addressing misconceptions about security threats has been discussed as important for

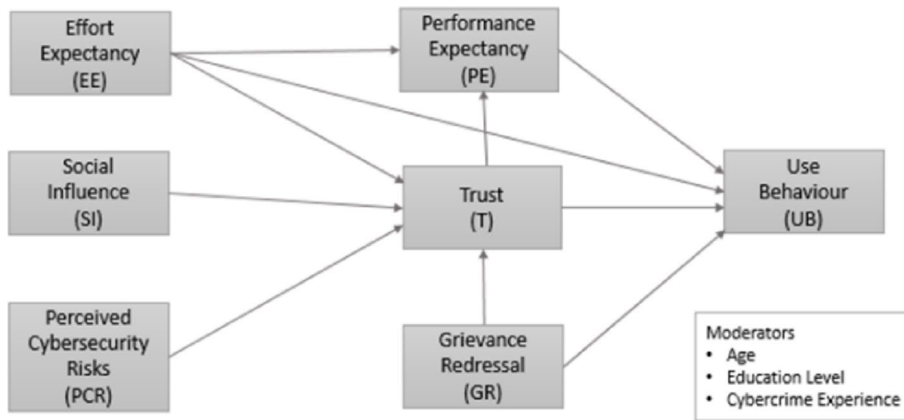

Fig. 1 The proposed research model

strengthening technology acceptance and satisfaction [33]. Consistent with this literature, higher perceived cybersecurity risk may be negatively associated with trust, given users' concerns about financial and personal data protection [36]. Studies confirm that elevated risks erode trust, which is essential for widespread use, as consumers prioritize data security over convenience [33]. Following the preceding discussion, the subsequent hypothesis posited:

H1 *Perceived cybersecurity risks will be negatively associated with trust towards digital payment systems.*

3.2 Social influence

Social Influence (SI) reflects the extent to which individuals perceive that significant others—such as friends, family, or colleagues—expect them to use a technology [37]. SI can shape users' beliefs through experiences and opinions shared within their social networks. Observing trust in a digital payment system among peers may strengthen an individual's trust and be associated with higher use [38].

Studies suggest that SI is often positively associated with trust by reinforcing perceptions of security, reliability, and widespread use [13, 39]. In Indian contexts, SI has also been reported as an important correlate of digital payment use behaviour [7, 33], though it showed no significant effect on mobile banking usage intentions [40]. To address this discrepancy, the second hypothesis is proposed:

H2 *Social influence will be positively associated with users' trust in digital payment systems.*

3.3 Effort expectancy

Effort expectancy (EE*)* denotes the 'level of simplicity and ease of use linked with the technology usage' [20]. Most research studies consider EE as one of the most significant determinants of a client's behavioural intention to use digital payment systems [7, 27, 41]. In digital payment contexts where security concerns are salient, a system's simplicity may reduce apprehension and be associated with more consistent use. Related work suggests that reducing the cognitive effort required to use digital payment platforms is associated with higher satisfaction and a greater likelihood of frequent use [42].

Accordingly, EE may be positively associated with trust, performance expectancy, and use behaviour. Therefore, we hypothesise:

H3 *Effort expectancy will be positively associated with users' trust in digital payment systems.*

H4 *Effort expectancy will be positively associated with users' use behaviour in digital payment systems.*

H5 *Effort expectancy will be positively associated with users' performance expectancy in digital payment systems.*

3.4 Grievance redressal

Grievance redressal (GR) refers to the established channels that enable clients to report and resolve concerns pertaining to their digital transactions [25]. Efficient mechanisms for resolving complaints can greatly improve client confidence and trust in digital payment platforms by immediately and fairly addressing their concerns, such as transaction problems or fraud [43].

Prior work in India discusses grievance redressal as an institutional mechanism that may be linked to usage outcomes [44]. When users have confidence in complaint resolution, they may be more willing to continue using digital payment platforms [25, 43]. On this basis, we propose:

H6 *Grievance redressal will be positively associated with users' trust in digital payment systems.*

H7 *Grievance redressal will be positively associated with users' use behaviour.*

3.5 Trust

Trust in digital payment systems reflects confidence in the service's integrity, benevolence, and proficiency [39] and the belief that transactions will meet expectations [45]. Trust is commonly treated as a key correlate of users' perceptions and technology use [46–48] and has been discussed as particularly relevant for continued use in digital services [49]. Prior research suggests that incorporating trust can improve the explanatory power of UTAUT-based models for behavioural intention [39].

Trust may also be associated with stronger performance related beliefs, particularly in settings where cybersecurity concerns are salient [50, 51]. While some studies report that trust is not always directly associated with mobile banking usage [45], its role in digital payment use behaviour directly or via performance expectancy remains theoretically relevant. Accordingly, we hypothesise:

H8 *Trust will be positively associated with performance expectancy of digital payment system users.*

H9 *Trust will be positively associated with users' use behaviour.*

3.6 Performance *expectancy*

Performance expectancy (PE), a key construct in the UTAUT model, reflects the belief that using a technology enhances performance outcomes [20] and relates to perceived efficiency and productivity benefits [52]. In digital payments, PE captures benefits such as improved task efficiency, reduced errors, and enhanced transaction convenience [42]. Prior studies frequently report PE as a strong correlate of behavioural intention and use [7, 26, 27], including evidence from Indian consumers and student samples [25, 43]. Based on this, the following hypotheses are proposed:

H10 *PE will be positively associated with digital payment use behaviour.*

H11 *PE mediates the relationship between effort expectancy and use behaviour.*

3.7 Cyb*ercrime experience*

Cybercrime in digital payment systems involves unlawful activities like unauthorized access, data theft, and fraudulent transactions [53]. Techniques such as phishing, malware, and scams exploit technological and behavioural vulnerabilities and have been discussed in relation to the growing use of digital payments [36]. Phishing and financial fraud are widely reported concerns in India [54, 55], and complaints logged via the National Cyber Crime Reporting Portal frequently involve digital payment contexts [56]. Prior research suggests that cybercrime experiences are often associated with higher perceived risk and lower trust [38, 55, 57]. Thus, we hypothesise that:

H12 *Cybercrime experience moderates the relationship between perceived cybersecurity risks and trust, such that the negative association between perceived risks and trust differs between users with and without cybercrime experience.*

3.8 Age and education level

Demographic characteristics can condition how users evaluate digital payment systems and, consequently, how strongly key antecedents relate to use behaviour. In the Indian digital payments context, evidence suggests that younger and more educated users are generally more inclined to engage with digital payment systems due to higher digital familiarity [12], while older and less digitally literate users often face barriers such as lower digital skills and heightened cybersecurity related concerns, underscoring the need to account for demographic heterogeneity in explaining use behaviour [58, 59]. Accordingly, we hypothesise that age and education level moderate the trust-formation pathways—specifically the effects of social influence, grievance redressal, and effort expectancy on trust—as stated in H13a–H14c.

H13a *Age moderates the relationship between social influence and trust, such that the SI→Trust association differs between younger and older users.*

H13b *Age moderates the relationship between grievance redressal and trust, such that the GR→Trust association differs between younger and older users.*

H13c *Age moderates the relationship between effort expectancy and trust, such that the EE→Trust association differs between younger and older users.*

H14a *Education level moderates the relationship between social influence and trust, such that the SI→Trust association differs between higher education and lower education users.*

H14b *Education level moderates the relationship between grievance redressal and trust, such that the GR→Trust association differs between higher education and lower education users.*

H14c *Education level moderates the relationship between effort expectancy and trust, such that the EE→Trust association differs between higher education and lower education users.*

4 Research methodology

4.1 Design of questionnaire and measures

The questionnaire used in this study was adapted from an instrument previously validated and published in our earlier work [60]. The instrument begins with questions on respondents' banking behaviours, such as the number of bank accounts, credit cards, and digital payment use behaviour, followed by 21 items measuring seven constructs, including performance expectancy, effort expectancy, trust, grievance redressal, social influence, perceived cybersecurity risk, and digital payment use behaviour using a 5-point Likert scale (see Appendix A). The cybercrime section includes scenario-based yes/no items addressing the ten most prevalent digital payment-related cybercrimes. Socioeconomic and demographic information were collected at the end of the questionnaire.

The questionnaire had a rigorous validation by experts following established guidelines [61, 62] and was reviewed by professors at IIT Kanpur to ensure contextual relevance. An online readability test was conducted with 120 participants (data not used in this study), and the instrument was translated into Hindi. A pilot study conducted at IIT Kanpur assessed internal consistency and clarity, leading to minor refinements in wording and flow.

4.2 Data collection

Data collection is crucial, as research conclusions depend on the quality of the data [63]. An initial online survey via social media introduced sample bias, favouring individuals with technological knowledge and stable internet access [64]. This was replaced with in-person surveys to ensure a diverse and representative sample, improving participation and allowing researchers to address participants' queries directly. Data were collected between March 2023 and March 2024 using a multi-stage purposive (non-probability) sampling strategy designed to ensure that respondents were informed and relevant participants, namely, individuals with direct experience in formal banking and potential exposure to digital payment systems. In the first stage, the sampling frame was defined to achieve broad regional representation across nine North Indian states and union territories, operationalised through 17 cities: Chandigarh, Delhi, Rohtak, Shimla, Jammu, Amritsar, Bikaner, Jaipur, Udaipur, Agra, Aligarh, Gorakhpur, Kanpur, Lucknow, Meerut, Varanasi, and Haldwani. These locations were chosen for their relevance within the evolving digital payment ecosystem, as well as for their logistical viability during the fieldwork period.

In the second stage, within each selected city, surveyors approached potential participants at designated bank branches after obtaining permission from branch managers. This recruitment approach was adopted to ensure authentic access to active banking populations and to minimise responses from individuals with limited relevance to the study context. Respondents were invited to participate only if they met the inclusion criterion of being active bank account holders, ensuring substantive relevance to digital payment use behaviour. The questionnaire was administered in English and Hindi, with trained field investigators available to clarify questions, thereby reducing response error and improving data quality.

Each city contributed approximately 50 valid responses, yielding 850 respondents overall, exceeding the required 210 based on the thumb rule [65].

4.3 Data screening

This study employed rigorous data screening and cleaning to ensure dataset integrity before analysis. The process involved identifying errors, inconsistencies, outliers, and missing data to meet the assumptions for robust analysis [61], reducing biases, and ensuring accurate results.

Respondents who were non-users of digital payment systems ($n = 18$; some incomplete), or who provided incomplete responses, were excluded. In addition, respondents reporting zero digital payment transactions in the last seven days were excluded because the study operationalises the dependent variable as recent use behaviour. The seven-day recall window was selected to minimise recall bias and ensure that responses reflect current, active use behaviour rather than sporadic or historical behaviour. A logical consistency check was also applied, and cases were removed when responses were found to be internally inconsistent. Missing data were addressed through list-wise deletion; no imputation techniques were employed. The values outside the valid 1–5 Likert range were considered as outliers and removed to prevent skewed estimates and distorted model fit [66]. Regarding field administration, non-responses were handled by approaching alternative eligible participants, however, refusal rates were not systematically logged across sites and are acknowledged as a limitation. Following these steps, a final dataset of 732 active participants was retained for analysis, as detailed in subsequent sections.

5 Analysis and findings

This study employed structural equation modelling (SEM), integrating regression and factor analysis, which allows for the examination of complex relationships among constructs while accounting for measurement error [67]. SEM is suitable for analysing associational relationships and potential mediating pathways in cross-sectional data. Following Anderson and Gerbing's two-step approach [68], the analysis included Confirmatory Factor Analysis (CFA) for the measurement model and path analysis for the structural model, ensuring reliable and comprehensive results [65].

5.1 Descriptive results

This study comprises a final sample of 732 responses. The predominant age group among respondents is Millennials, accounting for 61.9% of the total. Additionally, a substantial portion of the sample is highly educated, with 67.8% of respondents holding at least a Bachelor's degree. The survey results reveal that all participants had reported some

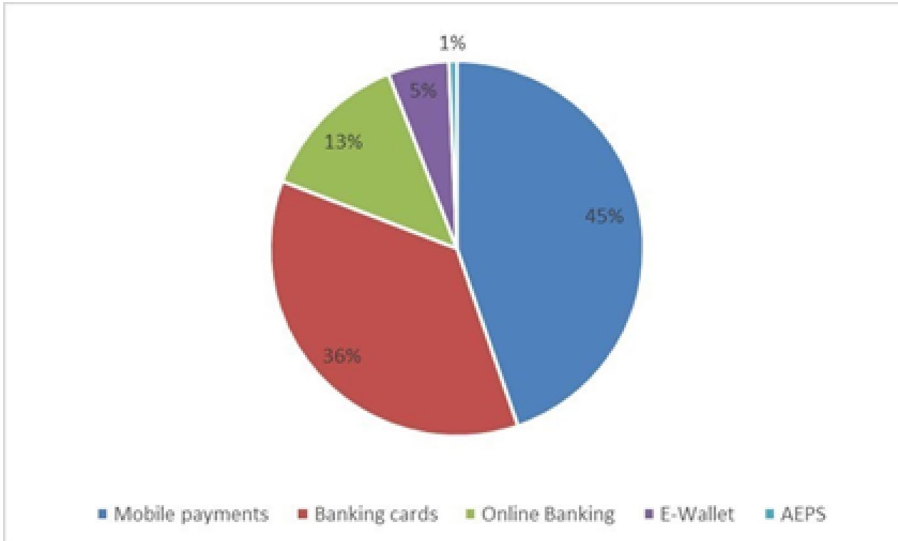

Fig. 2 Distribution of digital payment systems used in the last month

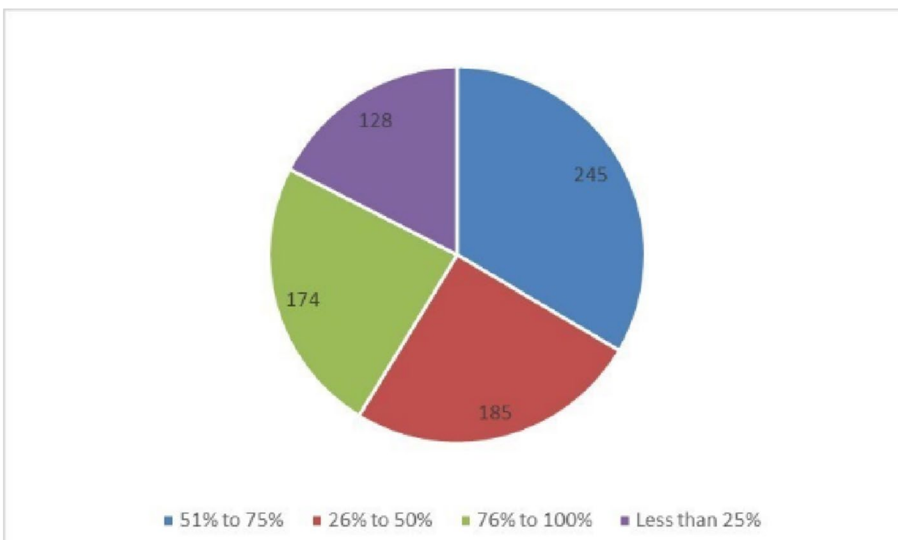

Fig. 3 Percentage of transactions via digital payments over the past week

level of digital payment use behaviour in the past month. Figure 2 illustrates that 45% of respondents utilised mobile payment platforms, rendering it the most preferred method. Subsequently, 36% of respondents employed banking cards, although 13% availed themselves of online banking services. Five percent of respondents utilised e-wallets, whereas only one percent reported employing AEPS services. Moreover, as illustrated in Fig. 3, the frequency of digital transactions over the past week confirms this trend, with the majority of participants (245 individuals) executing 51% to 75% of their transactions digitally, followed by 185 respondents performing 26% to 50% of their transactions through digital payment systems. These descriptive patterns indicate variability in digital payment use behaviour among participants.

Moreover, the survey examined respondents' experiences with cybercrime. The results reveal that a significant percentage of participants, specifically 608 respondents (around

83%), did not report any cybercrime incidents in the preceding year. The absence of reporting may be due to various factors, including insufficient awareness of cybercrime definitions, the belief that cybercrimes are too trivial to warrant reporting, or a lack of confidence in the efficacy of grievance redressal systems [69].

Conversely, 124 respondents (approximately 17%) indicated that they had encountered cybercrime. Notably, of those who have experienced cybercrime, 32 individuals have encountered multiple cybercrimes, underscoring the necessity for targeted interventions and enhanced cybersecurity measures to address the recurring nature of these experiences among affected individuals.

5.2 Measurement model

The measurement model was examined for convergent validity, discriminant validity, and internal consistency of the constructs through confirmatory factor analysis (CFA) [70]. Table 1 represents the CFA results, where standardized factor loadings (FL) quantify the intensity of the association between observed items and their corresponding latent constructs [65], average variance extracted (AVE) quantifies the variance accounted for by a latent construct with the overall variance [71], and composite reliability (CR) evaluates the internal consistency of construct indicators, akin to Cronbach's alpha [72]. The FL, CR, AVE, and alpha values surpass the thresholds of 0.7, 0.7, 0.5, and 0.7, respectively [73].

Table 1 Assessment results for the measurement model

Factor	FL	CR	AVE	α value
UB		0.842	0.642	0.83
UB1	0.848			
UB2	0.728			
UB3	0.822			
EE		0.841	0.571	0.83
EE1	0.775			
EE2	0.717			
EE3	0.811			
EE4	0.714			
SI		0.789	0.556	0.75
SI1	0.751			
SI2	0.690			
SI3	0.792			
T		0.917	0.789	0.90
T1	0.950			
T2	0.968			
T3	0.727			
GR		0.784	0.552	0.76
GR1	0.592			
GR2	0.823			
GR3	0.793			
PCR		0.901	0.819	0.89
PCR1	0.886			
PCR2	0.924			
PE		0.824	0.611	0.73
PE1	0.870			
PE2	0.759			
PE3	0.707			

Table 2 Fornell–Larcker criteria for discriminant validity

	CR	AVE	T	EE	SI	GR	PCR	PE	UB
T	0.917	0.789	**0.888**						
EE	0.841	0.571	0.390	**0.755**					
SI	0.789	0.556	0.369	0.659	**0.746**				
GR	0.784	0.552	−0.231	−0.363	−0.368	**0.743**			
PCR	0.901	0.819	−0.598	−0.249	−0.201	0.129	**0.905**		
PE	0.824	0.611	0.432	0.665	0.545	−0.298	−0.112	**0.782**	
UB	0.842	0.642	0.296	0.658	0.542	−0.124	−0.120	0.678	**0.801**

The bold diagonal values denote the square root of the AVE values

Table 3 HTMT results

	UB	GR	T	PE	SI	PCR	EE
UB							
GR	−0.10						
T	0.34	−0.28					
PE	0.78	−0.24	0.44				
SI	0.57	−0.30	0.30	0.60			
PCR	−0.15	0.24	−0.65	−0.10	−0.17		
EE	0.68	−0.33	0.47	0.59	0.55	−0.34	

The Fornell-Larcker test is used to evaluate that a construct with its indicators is distinct from other constructs by examining the square root of AVE of each construct in relation to its relationships with other factors [71]. Table 2 demonstrates that each factor had greater variance with its items than with those of other factors. Another rigorous approach to assessing the validity of discriminants is the Heterotrait-Monotrait (HTMT), which compares the average correlations between indicators across various factors with the average correlations within a single factor [74]. Table 3 illustrates sufficient discriminant validity; all HTMT values were below the 0.85 threshold.

The overall fit of the measurement model is good, with *CMIN/DF* = 2.82, *AGFI* = 0.919, *CFI* = 0.916, *RMSEA* = 0.050, and *PNFI* = 0.753, all of which fall within the acceptable thresholds [61, 75].

5.3 Structural model

SEM was performed using AMOS Graphics and SPSS Version 29.0. Before conducting the path analysis for the proposed model, it was essential to verify the model's fit indices. The results indicated a good fit (*CMIN/DF* = 2.70). Other key indices, including (*AGFI* = 0.916), (*CFI* = 0.957), (*RMSEA* = 0.052), and (*PNFI* = 0.768), were all within acceptable thresholds.

5.3.1 Hypothesis results

Structural relationships were evaluated using SEM, with standardised path coefficients, *t*-values, and *p*-values reported in Table 4. Overall, the model demonstrates substantial explanatory power, accounting for 44% of the variance in trust ($R^2 = 0.44$), 49% in performance expectancy ($R^2 = 0.49$), and 67% in use behaviour ($R^2 = 0.67$). These R^2 values indicate that the proposed model accounts for a substantial proportion of variance in the key endogenous constructs, supporting its relevance for understanding patterns of recent digital payment use behaviour within the sampled population.

Table 4 Hypothesis test results

Hypothesis	IC	DC	R^2	β	t-value	*P*-value	Support?
H1	PCR			−0.528	−15.203	***	Yes
H3	EE	Trust	0.44	0.156	3.201	**	Yes
H2	SI			0.137	2.909	*	Yes
H6	GR			−0.056	−1.513	0.130	No
H5	EE	Performance	0.49	0.601	11.266	***	Yes
H8	T	Expectancy		0.189	4.648	***	Yes
H7	GR			0.160	4.268	***	Yes
H4	EE	Use	0.67	0.339	6.340	***	Yes
H10	PE	Behavior		0.616	9.153	***	Yes
H9	T			−0.063	−1.635	0.102	No

***$P<0.001$, * $P<0.05$, ** $P<0.01$, IC, independent construct, DC, dependent

As shown in Table 4, PCR was negatively associated with trust ($\beta = -0.528$, $P<0.001$), supporting H1 and aligning with prior work emphasising the trust-eroding role of perceived security threats [33, 76]. SI was positively associated with trust ($\beta=0.137$, $P<0.05$), supporting H2 and indicating that social cues and peer endorsement are related to higher trust perceptions [38, 40]. EE also showed a positive association with trust ($\beta=0.156$, $P<0.01$), supporting H3, and was positively associated with UB ($\beta=0.339$, $P<0.001$), supporting H4, and aligning with previous studies [39, 45, 77, 78]. In addition, EE was strongly associated with PE ($\beta=0.601$, $P<0.001$), supporting H5, and suggesting that higher effort expectancy is linked with stronger performance expectations, consistent with prior findings [44, 79] findings.

Regarding GR, the hypothesised association with trust (H6) was not supported ($\beta = -0.056$, $P=0.130$). However, GR was positively associated with UB ($\beta=0.160$, $P<0.001$), supporting H7, and aligns with prior research [43, 44], indicating that perceptions of grievance redressal effectiveness are related to continued use behaviour even when a direct association with trust is not observed. Trust was positively associated with PE ($\beta=0.189$, $P<0.001$), supporting H8, whereas the direct association between trust and UB (H9) was not statistically supported ($\beta = -0.063$, $P=0.102$). Finally, PE exhibited the largest standardised association with UB ($\beta=0.616$, $P<0.001$), supporting H10 and underscoring the central role of perceived performance benefits in explaining recent use behaviour in this sample [7, 80]. Figure 4 provides a visual summary of the supported and unsupported hypotheses, including standardised path coefficients (β) and explained variance (R^2) for the endogenous constructs.

5.3.2 Mediation results

Mediation refers to the process by which a third variable, named the mediator, accounts for the statistical association between an independent variable and a dependent variable [81]. In mediation analysis, the independent variable is associated with the mediator, which in turn is associated with the dependent variable. In this study, mediation effects were evaluated in IBM SPSS AMOS v29 using maximum likelihood estimation and bootstrap resampling to obtain robust inference for indirect effects. Specifically, 5,000 bootstrap samples were generated and 95% bias-corrected confidence intervals (BC 95% CI) were computed. An indirect effect was considered statistically significant when its BC 95% CI did not include zero.

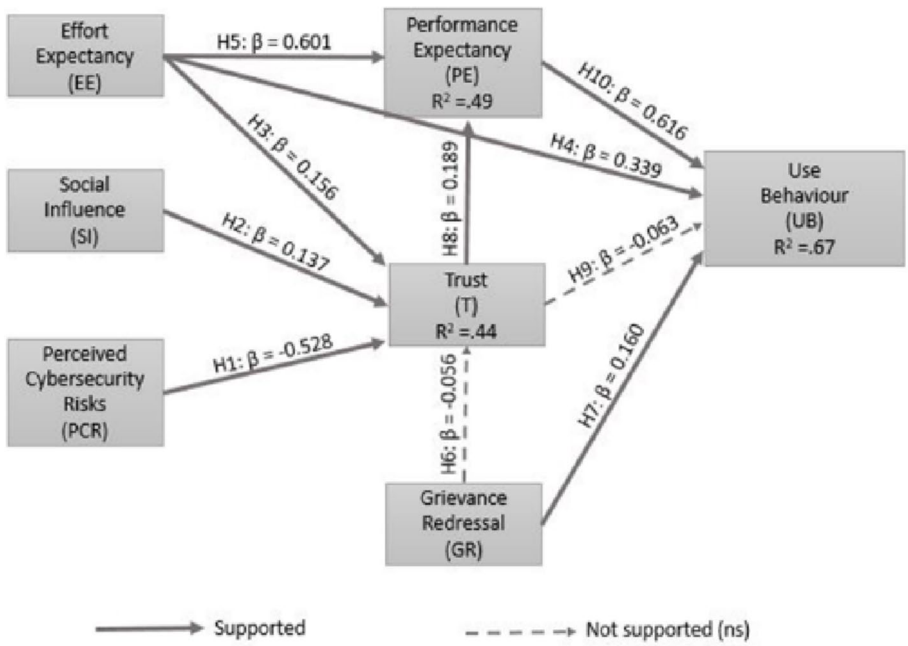

Fig. 4 Supported and unsupported hypotheses

Table 5 Mediation analysis results

Relationship	Direct effect	Indirect effect	Confidence interval		*P*-value	t-value	Conclusion
			Lower bound	Upper bound			
EE→PE→UB	0.404***	0.622	0.472	0.819	0.000	7.06	Partial mediation
EE→T→UB	0.404***	−0.019	−0.061	0.003	0.088	1.18	No mediation
PCR→T→UB	0.005 (Ns)	0.031	−0.011	0.073	0.144	1.47	No mediation
SI→T→UB	0.211**	−0.013	−0.047	0.002	0.086	1.08	No mediation
GR→T→UB	0.446***	0.010	−0.004	0.048	0.145	0.83	No mediation

Ns, not significant, ** $P<0.01$, *** $P<0.001$

H11 examined the mediation role of PE in the association between EE and UB. The findings demonstrated that the indirect effect of EE on UB was significant and positive (indirect effect $\beta = 0.622$, BC 95% CI [0.472, 0.819], $P = 0.000$), supporting hypothesis H11. However, the mediation was partial, as the direct effect of EE on UB remained substantial even with the mediator present (direct effect $\beta = 0.404$, $P < 0.001$). In contrast, the indirect effects of EE, PCR, SI, and GR on UB through Trust were not statistically significant because their BC 95% confidence intervals included zero and $P > 0.05$. Table 5 summarises the direct and indirect effects and the associated bootstrap confidence intervals.

5.3.3 Moderation results

Moderation occurs when the strength or direction of an association varies across subgroups [82]. To test whether key structural relationships vary across user characteristics, we conducted multi-group SEM in IBM SPSS AMOS v29 for age, education level (EL), and cybercrime experience (CE). Prior to interpreting cross-group differences in structural paths, we evaluated measurement invariance to confirm that the constructs are measured comparably across groups. Specifically, we estimated a configural model (same factor structure across groups,no equality constraints) followed by a metric invariance model (equal factor loadings). In each case, the metric invariance constraint did not significantly worsen model fit, supporting the validity of subsequent group comparisons. Given that the moderation analysis concerns structural path differences (rather than latent mean differences), establishing metric invariance was considered sufficient for interpretation.

Age. The sample was divided into Younger (≤ 30; n = 396) and Older (> 30; n = 336) groups. The configural multi-group model exhibited acceptable fit (CMIN/DF = 2.255, AGFI = 0.880, CFI = 0.947, PNFI = 0.741, RMSEA = 0.041). Metric invariance was supported when factor loadings were constrained equal across groups (ΔCMIN(14) = 14.342, P = 0.425). Structural comparisons indicate that trust-formation pathways differ by age, particularly for the associations of GR, SI, and EE with Trust, supporting H13a, H13b, and H13c. In the older group, EE was significantly associated with trust, whereas GR and SI were not statistically significant, indicating that perceived expectancy is more salient for trust formation among older users in this sample. In contrast, in the younger group, GR and SI were significantly associated with trust, while EE was not, consistent with the notion that younger users' greater baseline digital familiarity may reduce the importance of effort-related considerations in shaping trust [83]. Detailed group-specific estimates and between-group difference tests are reported in Appendix B, Table 6.

Cybercrime experience. The sample was divided into users with CE (n = 124) and without CE (n = 608). The multi-group model fit was acceptable (CMIN/DF = 2.639, AGFI = 0.901, CFI = 0.907, PNFI = 0.700, RMSEA = 0.054), and metric invariance was supported (ΔCMIN(14) = 12.546, P = 0.682). Results indicate that CE conditions the relative importance of institutional and usability cues; in the without-CE group, grievance redressal and effort expectancy show stronger associations with trust and use behaviour, whereas among users with CE, these pathways are attenuated and perceived cybersecurity risks becomes more salient, supporting H12. This pattern suggests that prior victimisation is associated with heightened sensitivity to cybersecurity risks relative to convenience-related considerations. Full multi-group estimates and difference tests are reported in Appendix B, Table 7.

Education level. The dataset was divided into High EL (bachelor's degree or higher; n = 496) and Low EL (schooling or less; n = 236) groups. The multi-group model fit was acceptable (CMIN/DF = 2.325, AGFI = 0.897, CFI = 0.954, PNFI = 0.752, RMSEA = 0.051), and metric invariance was supported (ΔCMIN(14) = 13.018, P = 0.605). The results indicate pronounced heterogeneity in trust formation across education levels. The association between SI and trust is statistically significant primarily among the highly educated group, whereas the associations of GR and EE with trust are more pronounced among the lower-educated group, supporting H14a, H14b, and H14c. This pattern suggests that users with lower education levels may rely more on institutional support mechanisms and perceived ease of use when forming trust in digital payment systems. Collectively, these findings underscore the importance of tailoring trust-building strategies to education-specific user needs. Detailed group-specific estimates and between-group difference tests are reported in Appendix B, Table 8.

6 Discussion and implications

The statistical findings confirm that the suggested model accounts for a substantial proportion of variance in use behaviour (67%), performance expectancy (49%), and trust (44%), meeting all measurement model criteria such as model fit, reliability, and validity. The study explores the associations of perceived cybersecurity risks, grievance redressal, trust, effort expectancy, performance expectancy, and social influence with digital payment system use behaviour in North India. While most hypotheses were supported, H9 and H6 were not.

Perceived cybersecurity risk exhibits a strong negative association with trust, reinforcing the centrality of security perceptions in shaping confidence in digital payment platforms. This suggests that even in contexts where digital payments are perceived as convenient, concerns about fraud, unauthorised access, and data breaches can undermine trust and be associated with lower reliance on these systems. In parallel, effort expectancy and social influence show positive associations with trust, indicating that perceptions of ease-of-use and normative endorsement from salient referents contribute to trust formation in digital payment environments. Performance expectancy emerges as the strongest direct associate of use behaviour, highlighting that perceived utility and efficiency gains remain primary correlates of use behaviour. Trust contributes indirectly to use behaviour through its positive association with performance expectancy, consistent with the interpretation that trust may strengthen perceptions of platform effectiveness rather than directly translating into higher near-term use behaviour.

The non-significant GR→Trust relationship (H6) suggests that perceived grievance efficiency does not necessarily generalise into trust for all users. One plausible explanation is that grievance redressal mechanisms are most salient when users encounter service failures or suspicious transactions. For individuals who have not experienced problems, have not reported them, or have not navigated formal complaint channels, grievance redressal may remain abstract and therefore weakly connected to trust judgments. This interpretation is consistent with the descriptive evidence that 608 respondents (≈83%) reported no cybercrime reporting in the preceding year, which may reflect limited awareness of cybercrime definitions, perceptions that incidents are not worth reporting, or low confidence in the efficacy of grievance redressal systems [69]. If a large share of users does not engage with grievance channels even when issues arise, grievance redressal becomes less visible as a trust-building mechanism at the population level, helping to explain the absence of a significant GR→Trust effect. Future research should explicitly measure awareness of reporting channels, prior complaint submission, and satisfaction with resolution, and model these factors as boundary conditions to assess whether grievance redressal is associated with higher trust primarily among users who have actively used these mechanisms.

Trust also does not exhibit a statistically significant direct association with use behaviour (H9). Two complementary explanations are plausible. First, within an active-user sample, trust may display restricted variance, making it harder to detect a direct effect on near-term use behaviour. Second, the results suggest that trust may influence use behaviour largely through instrumental beliefs, particularly performance expectancy, rather than operating as an independent behavioural driver. In contexts where users perceive strong functional benefits and manageable effort requirements, use behaviour may be determined more by perceived utility and convenience than by trust alone, especially in urban settings where digital payments are routinised.

The mediation analysis further supports these interpretations. Performance expectancy partially mediates the relationship between effort expectancy and use behaviour, indicating that effort expectancy is associated with higher use behaviour both directly and indirectly through stronger performance beliefs. By contrast, trust does not significantly mediate the remaining evaluated paths to use behaviour, reinforcing the view that trust operates more narrowly through performance perceptions and that other pathways may be contingent on user experience (e.g., engagement with grievance mechanisms).

The moderation results underscore meaningful heterogeneity in trust formation. Age-based comparisons indicate that the trust-formation role of effort expectancy and grievance redressal varies across younger and older users, suggesting that different cohorts rely on different cues when forming trust. Cybercrime experience further differentiates user priorities: for users without cybercrime experience, institutional assurance mechanisms and social cues are more salient correlates of trust, whereas for users with cybercrime experience, security concerns appear to dominate trust judgements, potentially attenuating the influence of usability or grievance mechanisms. Education level also conditions trust formation, with lower-educated users showing stronger reliance on grievance redressal and ease of use cues, while higher-educated users exhibit stronger sensitivity to social influence, plausibly reflecting differences in digital literacy, confidence, and informational networks [84]. Collectively, these patterns suggest that "one-size-fits-all" trust-building approaches are unlikely to be optimal, instead, trust in digital payment ecosystems is shaped by distinct pathways across demographic and experiential segments.

6.1 Theoretical implications

This study enhances the UTAUT framework by incorporating PCR, GR, and trust to better understand digital payment use behaviour, addressing gaps in traditional constructs like EE, PE, and SI. It highlights the salient association of PCR with trust and use behaviour, moderated by cybercrime experience, and emphasizes the role of GR in shaping trust perceptions, especially among users with lower education levels. Trust is further examined as a mechanism statistically linking PCR, SI, EE, and GR to use behaviour, offering insights into technology use in higher-risk environments, particularly in emerging economies. Importantly, these theoretical inferences should be interpreted within the study's sample boundary conditions, the analytic sample is drawn from urban regions and is skewed toward younger (millennial-dominant) and higher-educated active users. As a result, the strength and salience of the UTAUT pathways reported here may differ for older and lower-educated segments of the wider North Indian population, and further research using stratified sampling and explicit inclusion of under-represented groups is warranted.

6.2 Practical implications

The findings guide digital payment providers and policymakers in strengthening sustained use behaviour among active users in North India. Higher perceived cybersecurity risk is associated with lower trust, necessitating robust security measures, effective communication, and cybersecurity awareness campaigns. Simplifying interfaces and providing educational resources can improve accessibility for less tech-savvy users, with policymakers supporting digital literacy initiatives. GR, while not directly associated with trust, exhibits a positive association with use behaviour, underscoring the need for efficient grievance mechanisms and transparent complaint resolution frameworks. Highlighting performance benefits through marketing and fostering innovation with financial incentives can further support continued use. Social influence is pivotal, with peer advocacy and community-driven campaigns playing a key role. However, given that the sample is disproportionately younger and higher educated, these implications should not be treated as fully representative of the wider North Indian population. In practice, providers and policymakers should implement targeted interventions for older and lower-educated users such as assisted onboarding, local-language interfaces and simplified, low-literacy-friendly design, in-person helpdesks, and confidence-building security cues because these groups may rely more on usability support and institutional assurance than on peer influence alone.

6.3 Limitations

The findings should be interpreted in light of several limitations, which also point to opportunities for future research. First, the study relies on cross-sectional survey data; accordingly, the results indicate statistical associations rather than causal relationships. Longitudinal designs or experimental approaches would be valuable for establishing temporal ordering and testing causal mechanisms. Second, the analytic sample comprises active users, and the exclusion of non-users constrains generalisability with respect to barriers that prevent initial uptake of digital payment systems. Third, the screening criterion of zero digital payment transactions in the preceding seven days may under-represent infrequent users. Future work could adopt longer recall windows (e.g., 30 or 90 days), incorporate objective transaction logs where feasible, or model usage frequency explicitly. Finally, refusal and non-response rates were not systematically recorded across all survey sites, which may introduce unobserved selection effects. Subsequent studies should implement standardised logging of refusals and reasons for non-participation to strengthen the transparency and representativeness of the sampling process.

7 Conclusion and future research

This study examines factors associated with digital payment use behaviour in North India, with particular attention to perceived cybersecurity risks and trust. Extending the UTAUT framework with PCR, GR, and trust, the findings indicate that PE, EE, and GR are significantly associated with use behaviour, highlighting the salience of perceived performance benefits, ease of use, and accessible complaint-resolution mechanisms in explaining recent use patterns. PCR is negatively associated with trust, while EE and SI show positive associations with trust. Although trust does not exhibit a significant direct association with UB in this sample, it is positively associated with PE, indicating that trust may operate indirectly through performance related beliefs.

The moderation analyses further suggest that these associations are not uniform across users. Differences by age, education level, and cybercrime experience indicate heterogeneity in trust-formation pathways and use behaviour correlates, underscoring the importance of user segmentation in the design and governance of digital payment services. From a practical standpoint, the results support prioritising interventions that strengthen perceived performance benefits and usability while improving the visibility and accessibility of grievance mechanisms and reinforcing cybersecurity assurance and user awareness.

Future research may replicate and extend the model across additional Indian regions and comparable developing-economy contexts to assess geographic and cultural variation in these relationships. Additionally, interviews with policymakers and stakeholders could provide deeper insights into regulatory and operational challenges in the digital payment ecosystem.

Appendix A—Questionnaire

Construct	Code	Item	Source
Use behaviour (UB)	UB1	I pay for purchases using digital payment systems frequently	Zhou et al. [85]
	UB2	I use digital payment systems to manage my accounts	
	UB3	I use digital payment systems to do transactions	
Effort expectancy (EE)	EE1	Digital payment systems are simple to learn	Zhou et al. [85]; Venkatesh et al. [42]
	EE2	I find it easy to get digital payment systems to do what I want them to do	
	EE3	My interaction with digital payment systems is clear and understandable	
	EE4	I have the knowledge necessary to use digital payment systems	
Social influence (SI)	SI1	The use of digital payment systems is valued by my family and friends circle	Zhou et al. [85]; Venkatesh et al. [42]
	SI2	I find digital payment systems trendy and a status symbol in my society	
	SI3	The people around me influence me to use digital payment systems	
Trust (T)	T1	I believe digital payment systems are trustworthy	Zhou [86]; Srivastava et al. [87]; Lu et al. [88]
	T2	I trust digital payment systems to be secure	
	T3	Digital payment always provides accurate financial services	
Grievance redressal (GR)	GR1	In the case of failed digital payment transactions, there should be some authority to approach	Kumar et al. [43]
	GR2	There should be transparency in settling claims for failed digital payment transactions	
	GR3	Legal disputes about digital payments should be resolved in a timely manner	
Performance expectancy (PE)	PE1	Digital payment improves my performance in managing personal payments	Bhattacherjee [89, 90]; Zhou et al. [85]
	PE2	Digital payment systems help me advance my earnings	
	PE3	Using digital payment systems is time-saving	
Perceived cybersecurity risks (PCR)	PCR1	Internet hackers (criminals) might take control of my checking account if I used a digital payment system	Featherman and Pavlou [91]; Lallmahamood [92]
	PCR2	Using digital payment services subject your account to financial risk	

Appendix B

See Appendix Tables 6, 7, 8.

Table 6 Age moderation results

Relation	R^2	Older			Younger				Difference
		Estimate	t-value	*P*-value	R^2	Estimate	t-value	*P*-value	*P*-value
T←GR	0.47	0.051	0.957	0.339	0.43	0.150	2.888	**	*
T←PCR		−0.486	−9.612	***		−0.548	−11.437	***	ns
T←SI		0.134	1.810	0.070		0.126	2.524	*	*
T←EE		0.279	3.817	***		0.058	0.746	0.456	*
PE←T	0.62	0.135	2.262	*	0.40	0.227	4.035	***	ns
PE←EE		0.711	8.740	***		0.525	7.417	***	ns
UB←T	0.70	−0.078	−1.441	0.150	0.66	−0.062	−1.278	0.201	ns
UB←PE		0.539	4.713	***		0.660	7.877	***	ns
UB←GR		0.194	3.412	***		0.134	2.647	**	ns
UB←EE		0.456	4.227	***		0.288	4.284	***	ns

***$P < 0.001$, ns, not significant, ** $P < 0.01$, * $P < 0.05$

Table 7 Cybercrime experience moderation results

Relation	R^2	With CE			Without CE				Difference
		Estimate	t-value	*P*-value	R^2	Estimate	t-value	*P*-value	*P*-value
T←GR	0.42	0.025	0.280	0.779	0.45	0.083	2.024	*	*
T←PCR		−0.613	−6.364	***		−0.499	−13.230	**	**
T←SI		−0.015	−0.101	0.920		0.146	2.520	*	*
T←EE		0.257	1.785	0.074		0.170	3.053	**	*
PE←T	0.65	0.486	5.074	***	0.47	0.148	3.253	**	*
PE←EE		0.547	4.892	***		0.604	10.348	***	ns
UB←T UB←PE	0.57	−0.227	−1.375	0.169	0.69	−0.055	−1.475	0.155	ns
		0.822	2.678	**		0.571	8.524	***	ns
UB←GR		0.018	0.197	0.844		0.198	4.766	***	*
UB←EE		0.071	0.373	0.709		0.415	6.847	***	*

ns: not significant, ***$P < 0.001$, * $P < 0.05$, ** $P < 0.01$

Table 8 Education level moderation results

Relation	R^2	High EL			Low EL				Difference
		Estimate	t-value	*P*-value	R^2	Estimate	t-value	*P*-value	*P*-value
T←GR	0.44	0.009	0.192	0.848	0.47	0.174	2.787	**	*
T←PCR		−0.505	−11.683	***		−0.585	−9.962	***	ns
T←SI		0.186	2.617	**		0.002	0.021	0.983	*
T←EE		0.110	1.485	0.138		0.175	2.581	*	*
PE←T	0.49	0.240	4.319	***	0.44	0.181	2.684	**	ns
PE←EE		0.565	8.184	***		0.578	6.507	***	ns
UB←T UB←PE	0.65	−0.085	−1.720	0.085	0.63	−0.037	−0.613	0.540	ns
		0.626	6.473	***		0.576	5.686	***	ns
UB←GR		0.174	3.641	***		0.133	2.036	*	ns
UB←EE		0.328	4.353	***		0.352	3.860	***	ns

ns: not significant, ***$P < 0.001$, *$P < 0.05$, **$P < 0.01$

Author contributions

Conceptualization, Aya Aljaradat and Sandeep Shukla; methodology, Aya Aljaradat; software, Aya Aljaradat; validation, Aya Aljaradat and Sandeep Shukla; formal analysis, Aya Aljaradat; investigation, Aya Aljaradat; resources, Sandeep Shukla; data curation, Aya Aljaradat; writing—original draft preparation, Aya Aljaradat; writing—review and editing, Aya Aljaradat and Sandeep Shukla; visualization, Aya Aljaradat; supervision, Sandeep Shukla; project administration, Sandeep Shukla. All authors have read and agreed to the published version of the manuscript.

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Data availability
The datasets used in the study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Under the remit of the Institutional Ethics Committee (IEC), Indian Institute of Technology Kanpur (IITK), anonymised, non-interventional questionnaire studies of adult participants with no linked identifiers fall within the "exemption from review" category specified by the ICMR National Ethical Guidelines (2017, Table 4.2) for less-than-minimal-risk research. On this basis, formal IEC approval was not required. The study was carried out in accordance with the ethical standards of the IEC, the 1964 Declaration of Helsinki and its later amendments, and the ICMR guidelines. Informed consent was obtained from all individual participants included in the study.

Consent for publication
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Competing interests
The authors declare no competing interests.

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